Lime Harvesting Robot Challenge

This competition aims to encourage students to apply creativity and engineering skills in designing a functional robot. The task focuses on developing an automated system that can identify and pick ripe limes while leaving unripe ones on the tree. To ensure fairness, clear competition rules and technical specifications have been established for all teams. Evaluation will be conducted based on a structured marking rubric that combines robot functionality and task performance.

Figure 1: A plastic lime plant in a pot with artificial ripe (yellow) fruits, used as the target model for the robot-picking task.

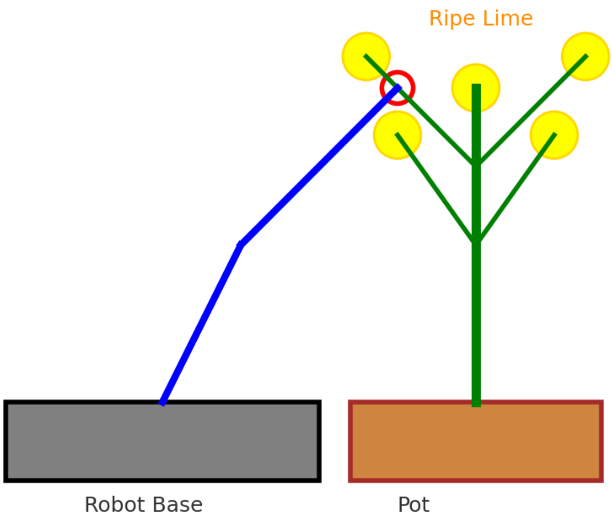

Figure 2: Conceptual illustration of the lime-picking robot. The robot base is positioned minimum 50 cm from the pot, with an arm and gripper mechanism reaching towards a ripe lime. *This is only a conceptual representation; the final design will depend on the creativity of each team.*

<u>Competition Rules</u>

1. **Robot Design**
   - Each team must design a robot capable of picking ripe limes (*see Figure 2*).
   - Ripe limes (yellow in color) must be picked and placed at the designated area.
   - Unripe limes (green in color) must remain on the tree.
2. **Plant and Robot Setup**
   - The base of the plant pot is positioned minimum 50 cm above the floor (*see Figure 1*).
   - The distance between the robot base and the pot is 50 cm.
   - The robot's working space must be at least 30 cm (width) × 30 cm (height) × 30 cm (length).
3. **Team Task**
   - Each team must pick **3 ripe limes**.
   - If unable to pick 3, then 2 or 1 ripe lime is acceptable.
4. **Winner Determination**
   - Evaluation will be based on two criteria:
     a) **Functionality of the Robot** – how well the robot performs the task (accuracy in picking ripe limes, ability to avoid unripe limes, and overall reliability of movement).
     b) **Task Performance** – number of ripe limes successfully picked. If the number is the same, the team that completes the task in the shortest time will be declared the winner.
5. **Overall Winner**
   - The overall winner will be determined by combining both criteria: **robot functionality** and **task performance**.

Competition Rubrics

Criteria	Excellent (8-10 points)	Good (6-7 points)	Fair (4-5 points)	Poor (0-3 point)
1. Fulfilling design specification				
Picking Success Rate (5)	Successfully picks 3 ripe limes.	Successfully picks 2 ripe limes.	Picks only 1 ripe lime.	Fails to pick any ripe lime.
Placement Success Rate (5)	Successfully places 3 ripe limes into designated area.	Successfully places 2 ripe limes into designated area.	Places only 1 ripe lime into designated area.	Fails to place any ripe lime.
2. Innovativeness				
Accuracy in Visual Servoing (10)	Demonstrates precise alignment between visual detection and end-effector movement during the attempt to pick 3 ripe limes.	Demonstrates precise alignment between visual detection and end-effector movement during the attempt to pick 2 out of 3 ripe limes.	Demonstrates precise alignment between visual detection and end-effector movement during the attempt to pick 1 out of 3 ripe limes.	Fails to demonstrate precise alignment between visual detection and end-effector.
3. Quality of work				
Design & Innovation (10**)**	Product shows excellent quality of work in: i. Human–Machine Interface (HMI) is **properly designed** with **user-friendliness** as a top priority. ii. The prototype is housed within a **well-designed enclosure or casing** that offers adequate **protection** and presents a **professional appearance**. iii. Wiring, piping and tubing (if applicable) are **neatly routed, professionally labelled and documented** in diagrams. iv. **Maximize** the utilization of **printed circuit board (PCB) or strip / donut board**. **Proper connectors** and **wires** are studied, selected and used.	Product shows good quality of work in: i. Human–Machine Interface (HMI) is **user-friendly**, but there may be **minor** areas where further **improvement** is possible. ii. The prototype is housed within a **well-designed enclosure or casing**, which may need a **minor improvement**. iii. Wiring, piping and tubing (if applicable) are **neatly routed, labelled and documented**, which may **require minor refinements** for greater clarity. iv. **Maximize** the utilization of **printed circuit board (PCB) or strip / donut board**. But the **connectors** and **wires** are used **without proper study**.	Product shows fair quality of work in: i. Human–Machine Interface (HMI) exhibits **some user-friendliness**. ii. The prototype is housed within an **enclosure or casing**, that **lacking** of either adequate **protection** functionality or profesional **appearance**. iii. Wiring, piping and tubing (if applicable) are **neatly routed, labelled and documented**, which may **require major refinements** for greater clarity. iv. The use of breadboard together with **printed circuit board (PCB) or strip / donut board**. **Jumper wires** are **mostly used**.	Product shows poor quality of work in: i. Human–Machine Interface (HMI) **lacks user-friendliness**. ii. The prototype is **not protected by an enclosure or casing**, that may affect the safety of the end-user and reliability of the prototype. iii. Wiring, piping and tubing (if applicable) are **disorganized** and **without a proper label or documentation**. iv. The use of **breadboard** and **jumper wires**.
4. Live demo				
Completion Time for Two Picking Attempts (10)	Completes the task within the given time.	Completes the task longer than the given time.	Manage to successfully pick and place at least 1 ripe.	Unable to complete the task.

Terms and conditions

1. Each team will be given **5 minutes** to complete the task, with a **maximum of two trials** allowed. The **total number of harvested lemons** from both trials will be **accumulated** for scoring.
2. The setup will consist of **three ripe** and **three unripe** lemons, which will be **placed randomly** during the competition.
3. The **stem** of each lemon will be made using **rolled brown paper** measuring approximately **1.5 cm × 3 cm**. Participants may select the **type of paper** based on **suitability** for their gripping or detection mechanism. Refer to Figures 3 and 4.
4. **Marks will be deducted by 50%** if a lemon is harvested **without proper cutting** of the stem.
5. The **judge's decision is final**.

Figure 3: The left-hand side shows the correctly rolled paper stem (the color should be brown). The right-hand side is incorrect, as the paper has not been rolled properly.

Figure 4: Multiple viewing angles of the lemons attached to the tree.

COMPONENTS LOAN FOR MECHATRONIC STUDENTS -CAPSTONE PROJECT

Raspberry Pi (Booked 10)	(9) Power supply 24V (NOT complete product) – Booked 3 need 3 more	(10) Stepper Motor NEMA 17 – Booked 3 need 3 more	(11) Stepper Motor NEMA 23 23HS22-1504S / 57BYGH56 – Booked 3 need 3 more
M2G07 M2G08 M1G13 M1G12 M1G08 M1G06 M2G13 M2G12 M2G11 M1G09	M1G11 M1G12 M2G01 M2G02 M2G03 M2G05	M1G11 M1G13 M2G01 M2G02 M2G04 M2G06	M1G11 M1G12 M2G01 M2G02 M2G03 M2G05

(12) Stepper Motor NEMA 23 23HS30-2804S1 – Booked 3 need 3 more	(13) Stepper Motor NEMA 23 57BYGH112 – Booked 3 need 3 more	(14) Stepper Motor Driver TB6600 – Booked 3 need 3 more	(15) Stepper Motor Driver DM542 -Booked 3 need 15 more. -1 group needs 3 units
M1G11 M1G13 M2G02 M2G03 M2G04 M2G06	M1G11 M1G12 M2G01 M2G03 M2G04 M2G05	M1G11 M1G13 M2G02 M2G04 M2G05 M2G06	M1G11 M1G12 M1G13 M2G01 M2G03 M2G05

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	A	B	C	D
1	**No**	**Component**	**Price (RM)**	**Stock**
2	1	Breadboard 16.5x5.5cm (830 Holes)	3.90	30
3	2	Solar Panel 5V USB Fixed, 5W	35.00	8
4	3	Solar Panel 18V 20W	85.00	8
5	4	Solar Charge Controller (10A)	18.00	6
6	5	AC Voltage Sensor Module 240V	6.90	0
7	6	Solar Power Bank 5V 2 USB-A 20000mAh	35.00	5
8	7	Power Supply 240Vac to 24Vdc+12Vdc+5Vdc (5A)	35.00	10
9	8	Power supply 5V and 12V (NOT complete product)	80.00	0
10	9	Power supply 24V (NOT complete product)	158.00	10
11	10	Stepper Motor NEMA 17	40.00	10
12	11	Stepper Motor NEMA 23 23HS22-1504S / 57BYGH56	60.00	8
13	12	Stepper Motor NEMA 23 23HS30-2804S1	117.00	8
14	13	Stepper Motor NEMA 23 57BYGH112	117.00	8
15	14	Stepper Motor Driver TB6600	30.00	10
16	15	Stepper Motor Driver DM542 3 unit	60.00	24